

Lecture 24

Linking Reinforcement Learning and Evolutionary Dynamics

COMP532 – Machine Learning and Bioinspired Optimisation

Meng Fang, University of Liverpool



Today's Questions

At the end of this lecture, you should be able to answer:

1. How can learning be viewed as a dynamical process?
2. What is the connection between reinforcement learning updates and evolutionary dynamics?
3. Why is this connection useful for understanding multi-agent learning?
4. What are the limits of the analogy between RL and EGT?

Recap of Lecture 23

In the previous lecture, we introduced:

- ▶ the population view of strategic behaviour
- ▶ payoff as relative fitness
- ▶ replicator dynamics
- ▶ evolutionarily stable strategies (ESS)
- ▶ qualitative behaviours such as convergence, coexistence, and cycling

Today's step

We now ask whether **learning algorithms** can also be understood as generating structured dynamics over time.

From Population Dynamics to Learning Dynamics

Evolutionary game theory studies how **strategy frequencies** change over time.

Reinforcement learning studies how **policies, values, or preferences** change over time through interaction.

In both cases, we have:

- ▶ adaptation
- ▶ feedback from rewards or payoffs
- ▶ trajectories over time
- ▶ possible convergence, instability, or oscillation

Key idea

Both EGT and RL are concerned not only with *what* is good, but with *how behaviour changes*.

A Common Structure

At a high level, both reinforcement learning and evolutionary dynamics follow the same pattern:

1. start from a current strategy or behaviour
2. interact with the environment or other agents
3. receive feedback in the form of payoff or reward
4. update behaviour and repeat over time

Key point

Both can be viewed as systems that repeatedly adapt behaviour based on feedback.

A Dictionary Across Frameworks

Reinforcement Learning	Classical Game Theory	Evolutionary Game Theory
environment	game	game
agent	player	population
action	action	type
policy	strategy	distribution over types
reward	payoff	fitness
update rule	best response / equilibrium reasoning	replicator dynamics

Key insight

These frameworks differ in interpretation, but share a common **dynamic structure**.

Learning as Motion in Policy Space

In reinforcement learning, a policy is not just a final answer. It is a **time-evolving object** that changes during training.

For example, if a policy over actions is parameterised by probabilities

$$\pi(a) = (\pi_1, \dots, \pi_k),$$

then learning induces a trajectory:

$$\pi^{(0)} \rightarrow \pi^{(1)} \rightarrow \pi^{(2)} \rightarrow \dots$$

Key shift

Instead of asking only “where does learning end?”, we can ask: **what path does it follow?**

Replicator-Like Behaviour in Policy Updates

Recall the replicator dynamic:

$$\dot{x}_i = x_i(f_i(x) - \bar{f}(x)).$$

At an abstract level, some policy update rules behave similarly:

- ▶ current probability matters
- ▶ advantage matters
- ▶ update direction is shaped by relative performance

Interpretation

The policy update amplifies actions whose estimated value exceeds a baseline.

This is one reason evolutionary dynamics can be a useful lens for analysing RL.

A Dynamic Lens on MARL Algorithms

Using the ideas from this lecture, we can reinterpret MARL methods as inducing different types of dynamics.

For example:

- ▶ independent learners may create unstable moving-target dynamics
- ▶ equilibrium-based methods may impose stronger structure but incur heavier computation
- ▶ gradient methods may exhibit smooth adaptation, cycling, or sensitivity to step size

Takeaway

Different algorithms are not just different optimisers. They are different **dynamical systems**.

Limits of the Analogy

It is important not to overstate the link.

RL and EGT are related, but not identical:

- ▶ RL often involves explicit state transitions and delayed reward
- ▶ function approximation can create high-dimensional, nonlinear dynamics
- ▶ exploration, replay, and optimisation heuristics complicate analysis
- ▶ real MARL systems may be partially observed and non-ideal

Balanced view

Evolutionary dynamics provides a useful **lens**, not a perfect one-to-one identity.

What This Means Methodologically

When studying a multi-agent learning system, we should ask not only:

- ▶ What objective is being optimised?

but also:

- ▶ What update rule is being used?
- ▶ What feedback loops does it create?
- ▶ What trajectories does it induce?
- ▶ Is the resulting behaviour stable, robust, and interpretable?

Methodological shift

Evaluation should include **dynamics**, not only final performance.

A Mental Model to Keep

- ▶ Game theory: **What strategic outcomes are possible?**
- ▶ EGT: **How do behavioural proportions evolve?**
- ▶ RL / MARL: **How do policies adapt through feedback?**

This lecture's core message:

Learning algorithms can often be understood as dynamical systems shaped by reward feedback and interaction structure.

Summary

1. Reinforcement learning can be viewed as a dynamical process in policy or parameter space.
2. Evolutionary dynamics provides useful concepts for understanding such trajectories.
3. The RL-EGT connection is useful for analysis, comparison, and algorithm design.
4. The connection is insightful, but not exact.

Looking Ahead

Next lecture:

- ▶ Why do people use multiple LLM agents instead of one?
- ▶ What kinds of coordination and communication problems arise in LLM multi-agent systems?
- ▶ How can MAL, game theory, and dynamics help us reason about them?

Bridge to Lecture 25

Modern LLM multi-agent systems are a natural setting where **interaction**, **adaptation**, and **stability** all matter at once.